Room Number ________

Seat Number ________

Student Number |__|__|__|__|__|__|__|__|__|__|

ANONYMOUSLY MARKED

(Please do not write your name on this exam paper)

CONFIDENTIAL EXAM PAPER

This paper is not to be removed from the exam venue

Mathematics and Statistics

EXAMINATION

Semester 1 - Main, 2018

MATH1921/MATH1931 Calculus Of One Variable (Advanced)/(SSP)

For Examiner Use Only

Q	Mark
1	
2	
3	
4	

EXAM WRITING TIME: 1.5 hours

READING TIME: 10 minutes

EXAM CONDITIONS:
This is a CLOSED book examination - no material permitted

During reading time - writing is not permitted at all

MATERIALS PERMITTED IN THE EXAM VENUE:
(No electronic aids are permitted e.g. laptops, phones)

Calculator - non-programmable

MATERIALS TO BE SUPPLIED TO STUDENTS:
Dept-specific answer material - provided by dept

INSTRUCTIONS TO STUDENTS:
This examination has two sections: Multiple Choice and Extended Answer.

The Multiple Choice Section is worth 30% of the total examination; there are 15 questions; the questions are of equal value; all questions may be attempted. Answers to the Multiple Choice questions must be entered on the Multiple Choice Answer Sheet.

The Extended Answer Section is worth 70% of the total examination; there are 4 questions; the questions are of equal value; all questions may be attempted; working must be shown.

A list of standard derivatives and integrals is provided at the back of the booklet.

Please tick the box to confirm that your examination paper is complete. ☐

Extended Answer Section

*There are **four** questions in this section, each with a number of parts.*
Write the answers in the booklets provided.

1. (a) (*i*) Write the complex number $z = -1 - \sqrt{3}i$ in modulus–argument form.

(*ii*) Write $3e^{\frac{\pi}{3}i}e^{\frac{\pi}{6}i}$ in Cartesian form.

(b) Determine whether or not the function $f\colon \mathbb{R} \to \mathbb{C}$ given by $f(t) = te^{it}$ is injective.

(c) Let A be the line $\operatorname{Im} z = -\dfrac{1}{2}$ in the complex plane with $-1 \leq \operatorname{Re} z < 2$.

(*i*) Sketch the set A on the complex plane.

(*ii*) Sketch the image of A under the map $z \mapsto z^2$ on the complex plane.
Carefully label the x-intercepts and the endpoints.

2. (a) Find the 4th order Taylor polynomial of $f(x) = \sqrt{1 + 4x^2}$ centred at $x = 0$.

(b) Is it possible to construct a function $f\colon [1, 3] \to \mathbb{R}$ that is continuous on $[1, 3]$, differentiable on $(1, 3)$ and for which $f(1) = -1$, $f(3) = 4$ and $f'(x) \leq 2$ for all $x \in (1, 3)$? Provide some explanation for your answer.

(c) Compute $\displaystyle\lim_{x\to\infty} \left(\frac{x^2}{x+3} - \frac{x^2}{x-3} \right)$.

3. (a) Compute $\dfrac{d}{dx}\displaystyle\int_0^{\sqrt{x}} \frac{t^2}{1+t^4}\,dt$ given that $x > 0$.

(b) Compute the definite integral $\displaystyle\int_0^1 \frac{x-2}{(x+2)(x+3)}\,dx$.

(c) (i) Let $A > 0$ be a constant and let $n, m \in \mathbb{N}$ so that $n > m$. Show that

$$\frac{A^n}{n!} \leq \frac{A^m}{m!}\left(\frac{A}{m+1}\right)^{n-m}.$$

(ii) Hence, or otherwise, show that $\dfrac{A^n}{n!} \to 0$ as $n \to \infty$ for every constant $A > 0$.

(iii) By estimating the remainder term in the n-th order Taylor polynomial of $f(x) = e^x$, prove that

$$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}$$

for all $x \in \mathbb{R}$.

4. (a) Compute the indefinite integral $\displaystyle\int \tan^3 x\,dx$.

(b) Determine whether or not the improper integral $\displaystyle\int_0^{\infty} \frac{\sin x}{x^{3/2}}\,dx$ exists.

(c) Assume that $[a, b]$ is a closed bounded interval and that $f\colon [a, b] \to \mathbb{R}$ is continuous and not constant. Use the intermediate value theorem and the extreme value theorem to prove that the image $\operatorname{im}(f) := \{f(x) \mid x \in [a, b]\}$ of f is a closed and bounded interval.

Table of Standard Integrals

1. $\displaystyle\int x^n\,dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1)$

2. $\displaystyle\int \frac{dx}{x} = \ln|x| + C$

3. $\displaystyle\int e^x\,dx = e^x + C$

4. $\displaystyle\int \sin x\,dx = -\cos x + C$

5. $\displaystyle\int \cos x\,dx = \sin x + C$

6. $\displaystyle\int \tan x\,dx = -\ln\left|\cos x\right| + C$

7. $\displaystyle\int \cot x\,dx = \ln\left|\sin x\right| + C$

8. $\displaystyle\int \frac{dx}{a^2+x^2} = \frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right) + C$

9. $\displaystyle\int \sec^2 x\,dx = \tan x + C$

10. $\displaystyle\int \operatorname{cosec}^2 x\,dx = -\cot x + C$

11. $\displaystyle\int \sec x\,dx = \ln\left|\sec x + \tan x\right| + C$

12. $\displaystyle\int \operatorname{cosec} x\,dx = \ln\left|\operatorname{cosec} x - \cot x\right| + C$

13. $\displaystyle\int \sinh x\,dx = \cosh x + C$

14. $\displaystyle\int \cosh x\,dx = \sinh x + C$

15. $\displaystyle\int \tanh x\,dx = \ln\cosh x + C$

16. $\displaystyle\int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1}\left(\frac{x}{a}\right) + C \qquad (|x| < a)$

17. $\displaystyle\int \frac{dx}{\sqrt{x^2+a^2}} = \sinh^{-1}\left(\frac{x}{a}\right) + C = \ln\left(x + \sqrt{x^2+a^2}\right) + C'$

18. $\displaystyle\int \frac{dx}{\sqrt{x^2-a^2}} = \cosh^{-1}\left(\frac{x}{a}\right) + C = \ln\left(x + \sqrt{x^2-a^2}\right) + C' \quad (x > a)$

Linearity: $\displaystyle\int \big(\lambda f(x) + \mu g(x)\big)\,dx = \lambda\int f(x)\,dx + \mu\int g(x)\,dx$

Integration by substitution: $\displaystyle\int f(u(x))\frac{du}{dx}\,dx = \int f(u)\,du$

Integration by parts: $\displaystyle\int f(x)g'(x)\,dx = f(x)g(x) - \int f'(x)g(x)\,dx$

End of Extended Answer Section

END OF EXAMINATION